ELG 3125 - Signals & Systems

Lecture 5 - Part 1

Convolution integral

Representation of Continuous-Time Signals in Terms of Impulses

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$$ ⇒ analogous to sifting property from Lecture 4.1 ⇒ $$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Similar to discrete signals ⇒ the response to the input signal $x(t)$ can be written as a convolution integral:

⇒ $$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$ ⇒ $$y(t) = x(t) * h(t)$$

Same logic is used to calculate convolution sums will be used to calculate convolution integrals.

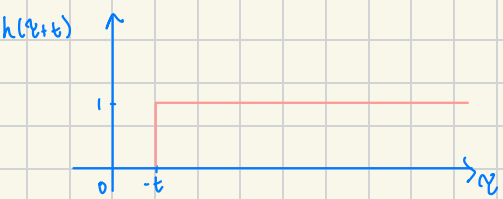
Example: $x(t) = e^{-at}u(t)$, $a>0$ and $h(t) = u(t)$

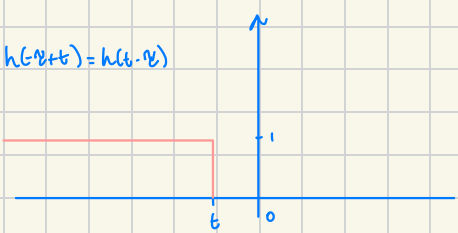


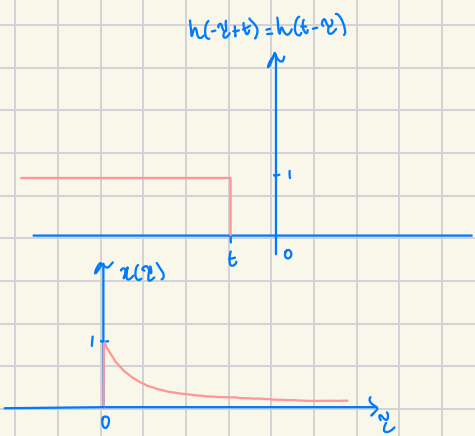


$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$

$h(t-\tau) = h(-\tau+t)$







For $t \le 0$: no overlap $\Rightarrow$ $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau = 0$

For $t \ge 0$: overlap region : from 0 to t $\rightarrow$ $y(t) = \int_0^t x(\tau)h(t-\tau)\,d\tau$

$y(t) = \int_0^t x(\tau)h(t-\tau)\,d\tau = \int_0^t e^{-a\tau}\underbrace{u(\tau)}_{=1}\underbrace{h(t-\tau)}_{=1}\,d\tau = \int_0^t e^{-a\tau}\,d\tau$, $[-a\tau]' = -a \Rightarrow 1 = -\frac{1}{a}[-a\tau]'$

$y(t) = \int_0^t -\frac{1}{a}[-a\tau]' e^{-a\tau}\,d\tau = -\frac{1}{a}\int_0^t [-a\tau]' e^{-a\tau}\,d\tau = -\frac{1}{a}\left[e^{-a\tau}\right]_0^t = -\frac{1}{a}\left[e^{-at} - e^{-a(0)}\right] = -\frac{1}{a}\left[e^{-at} - 1\right] = \frac{1}{a}\left(1 - e^{-at}\right)$

$\Rightarrow$ $y(t) = \frac{1}{a}\left(1 - e^{-at}\right)$

Overall output

$\therefore$ $y(t) = \begin{cases} 0, & \text{for } t < 0 \\ \frac{1}{a}\left(1 - e^{-at}\right), & \text{for } t \ge 0 \end{cases} = \frac{1}{a}\left(1 - e^{-at}\right)u(t)$

Overall Graph

